

Engineering Physics Lab

Engineering Physics Lab is the core of innovation in Engineering Physics, and the main resource for students to learn the technical and leadership aspects of becoming professional engineers. With their unique combination of engineering breadth and deep analytic training, students are exceptionally well prepared to help bring discoveries from the lab into real-world innovations.

The objective of the laboratory is skill development. The experiments are designed to illustrate phenomena in different areas of Physics and to expose students to measuring instruments. An Engineering Physics degree prepares students to work in the private sector or in national laboratories at the very forefront of technology, or to pursue an advanced degree in engineering.

Engineering Physics labs include experiments from Physics I and II. Physics Explorations and Projects is a collection of laboratory investigations aligned to the Next Generation Science Standards (NGSS). Most included investigations set up a situation for students to explore and analyze with some guidance from the instructor—a guided-inquiry format that is very different from traditional laboratory.

List of Experiments

1. **Milgram Experiment:** A series of social psychology experiments conducted by Stanley Milgram in the 1960s, which measured the willingness of study participants to obey an authority figure who instructed them to perform acts that conflicted with their personal conscience.
2. **Stanford Prison Experiment:** A social psychology experiment conducted by Philip Zimbardo in 1971, which aimed to investigate the psychological effects of perceived power, focusing on the struggle between prisoners and prison officers.
3. **Pavlov's Dogs:** A series of experiments conducted by Ivan Pavlov in the 1890s, which investigated classical conditioning, a type of learning in which an organism comes to associate stimuli.
4. **Asch Conformity Experiment:** A series of studies conducted by Solomon Asch in the 1950s, which demonstrated the power of conformity in groups and the influence of majority opinion on individual judgment.
5. **Hawthorne Effect:** A type of reactivity in which individuals modify or improve an aspect of their behavior in response to their awareness of being observed, first described in a series of experiments conducted at the Hawthorne Works factory in the 1920s and 1930s.

Any ten experiments (at least four from each group)

Here are ten experiments, with at least four from each group:

Group 1: 1. The Milgram Experiment: This experiment, conducted by Stanley Milgram in 1961, aimed to measure the willingness of participants to

obey an authority figure who instructed them to perform acts that conflicted with their personal conscience. 2. **The Stanford Prison Experiment:** This experiment, conducted by Philip Zimbardo in 1971, aimed to investigate the psychological effects of perceived power, focusing on the struggle between prisoners and prison guards. 3. **The Asch Conformity Experiment:** This experiment, conducted by Solomon Asch in 1951, aimed to investigate the extent to which social pressure from a majority group could affect a person to conform. 4. **The Bobo Doll Experiment:** This experiment, conducted by Albert Bandura in 1961, aimed to investigate if social behaviors (i.e., aggression) can be acquired by observation and imitation.

Group 2: 1. **The Double-Slit Experiment:** This experiment, first performed by Thomas Young in 1801, aimed to demonstrate the wave-particle duality of light. 2. **The Michelson-Morley Experiment:** This experiment, conducted by Albert Michelson and Edward Morley in 1887, aimed to detect the presence of the luminiferous aether, a supposed medium for the propagation of light. 3. **The Rutherford Gold Foil Experiment:** This experiment, conducted by Ernest Rutherford in 1909, aimed to investigate the structure of the atom. 4. **The Cavendish Experiment:** This experiment, conducted by Henry Cavendish in 1798, aimed to measure the force of gravity between masses and determine the gravitational constant.

Group 3: 1. **The Pavlov's Dog Experiment:** This experiment, conducted by Ivan Pavlov in the 1890s, aimed to investigate classical conditioning, a type of learning in which an organism comes to associate stimuli. 2. **The Skinner Box Experiment:** This experiment, conducted by B.F. Skinner in the 1930s, aimed to investigate operant conditioning, a type of learning in which behavior is strengthened or weakened by the use of reinforcement or punishment.

Group A

Group A is a term that can refer to many different things depending on the context. Here are some possible interpretations of the term "Group A":

1. In the context of sports, Group A may refer to a specific group or division within a tournament or league.
2. In the context of biology, Group A may refer to a specific classification of organisms.
3. In the context of chemistry, Group A may refer to a specific group of elements in the periodic table.
4. In the context of business, Group A may refer to a specific group of companies or a specific business unit within a larger organization.

1. To determine the wavelength of sodium light by Newton's ring experiment.

- Newton's ring experiment is a method to determine the wavelength of light using the phenomenon of interference.

- In this experiment, a plano-convex lens is placed on a flat glass plate, and a monochromatic light source, such as sodium light, is used to illuminate the setup.
- The light waves reflected from the top and bottom surfaces of the air film between the lens and the glass plate interfere with each other, producing a pattern of bright and dark rings known as Newton's rings.
- The radius of the rings can be measured using a traveling microscope, and the wavelength of the light can be calculated using the formula: $\lambda = (D_n^2 - D_{n-1}^2) / 4R$, where λ is the wavelength of the light, D_n is the diameter of the n th dark ring, R is the radius of curvature of the lens, and n is the order of the ring.
- This experiment is named after Sir Isaac Newton, who first described the phenomenon in his book "Opticks" in 1704.

2. To determine the wavelength of different spectral lines of mercury light using plane

- Mercury light is a type of light source that emits light when an electric current is passed through mercury vapor.
- The light emitted by mercury light contains several distinct spectral lines, each corresponding to a specific wavelength of light.
- To determine the wavelength of these spectral lines, a device called a spectrometer can be used.
- A spectrometer works by separating light into its constituent wavelengths using a diffraction grating or a prism.
- The light is then focused onto a detector, which measures the intensity of light at each wavelength.
- By analyzing the intensity of light at each wavelength, the wavelengths of the different spectral lines of mercury light can be determined.
- This information can be used for various applications, such as calibrating other instruments or studying the properties of mercury vapor.

Transmission Grating

- Transmission gratings are designed to disperse incident light at specific angles. Angular dispersion is a function of angle of incidence and groove spacing. Dispersion increases as angle of incidence increases or as groove spacing decreases.
- Transmission gratings are manufactured in the same way as ruled reflection gratings in which a replica of a master grating is created by duplicating its grooved surface with a thin, vacuum deposition-coated layer, except they have no reflective overcoat.
- Transmission gratings offer low polarization sensitivity as opposed to reflection gratings because incident light is not reflected by a mirror coating such as aluminum. Since light is transmitted through the grating, transmission gratings can be used in compact, in-line configurations.

- Transmission gratings are diffraction gratings which are used in transmission. This is in contrast to the more common reflection gratings, which reflect all incident light, as far as it is not absorbed or scattered. Transmission gratings can be volume Bragg gratings or surface relief gratings.
- The product line of transmission gratings by Coherent utilizes state-of-the-art design and precise lithographic wafer-scale fabrication to deliver record-breaking optical performance.

3. To determine the specific rotation of cane sugar solution using polarimeter.

- A polarimeter is an instrument used to measure the angle of rotation of plane-polarized light as it passes through an optically active substance.
- Specific rotation is a property of a substance that describes the degree to which the substance is able to rotate plane-polarized light.
- To determine the specific rotation of a cane sugar solution using a polarimeter, the following steps can be followed:
 1. Prepare a solution of cane sugar by dissolving a known amount of cane sugar in a known volume of water.
 2. Fill a polarimeter tube with the cane sugar solution.
 3. Place the polarimeter tube in the polarimeter.
 4. Adjust the polarimeter to obtain a reading of the angle of rotation of the plane-polarized light as it passes through the cane sugar solution.
 5. Use the observed angle of rotation, the concentration of the cane sugar solution, and the length of the polarimeter tube to calculate the specific rotation of the cane sugar solution using the formula: $[\alpha] = \alpha / (l * c)$, where $[\alpha]$ is the specific rotation, α is the observed angle of rotation, l is the length of the polarimeter tube, and c is the concentration of the cane sugar solution.
- The specific rotation of the cane sugar solution can provide information about the purity and concentration of the cane sugar in the solution.

4. To determine the focal length of the combination of two lenses separated by a distance

1. The focal length of the combination of two lenses separated by a distance can be determined using the lens formula.
2. The lens formula is given by $1/f = 1/v - 1/u$, where f is the focal length of the lens, v is the distance of the image from the lens, and u is the distance of the object from the lens.
3. To determine the focal length of the combination of two lenses, the distance between the lenses must be known.
4. The distance between the lenses can be measured using a ruler or a vernier caliper.
5. Once the distance between the lenses is known, the focal length of the combination can be calculated using the formula $1/F = 1/f_1 + 1/f_2$ -

d/f_1f_2 , where F is the focal length of the combination, f_1 and f_2 are the focal lengths of the individual lenses, and d is the distance between the lenses.

6. This formula can be used to calculate the focal length of the combination of two lenses separated by a distance. It is important to note that the sign conventions must be followed while using this formula.
7. The focal length of the combination of two lenses can also be determined experimentally by measuring the distance of the image and the object from the lens combination and using the lens formula to calculate the focal length.

Focal Length of Combination of Lenses

The formula for the focal length of a combination of lenses is given by the formula:

$$1/f = 1/f_1 + 1/f_2 - d/f_1f_2$$

where f is the focal length of the combination of lenses, f_1 and f_2 are the focal lengths of the individual lenses, and d is the distance between the two lenses.

To verify this formula, we can use the lens maker's formula, which states that the power of a lens is given by the formula:

$$P = 1/f = (n-1)(1/R_1 - 1/R_2)$$

where n is the refractive index of the lens material, R_1 and R_2 are the radii of curvature of the two surfaces of the lens, and f is the focal length of the lens.

For a combination of two lenses, the total power of the combination is given by the sum of the powers of the individual lenses. Therefore, we have:

$$P = P_1 + P_2$$

Substituting the lens maker's formula for the power of each lens, we get:

$$P = (n_1-1)(1/R_{11} - 1/R_{12}) + (n_2-1)(1/R_{21} - 1/R_{22})$$

where n_1 and n_2 are the refractive indices of the two lenses, R_{11} and R_{12} are the radii of curvature of the first lens, and R_{21} and R_{22} are the radii of curvature of the second lens.

Since the power of a lens is equal to the reciprocal of its focal length, we can write the above equation in terms of the focal lengths of the individual lenses:

$$1/f = 1/f_1 + 1/f_2$$

However, this formula assumes that the two lenses are in contact. If there is a distance d between the two lenses, we need to account for the effect of this distance on the total power of the combination.

To do this, we can use the formula for the power of a thin lens in air, which states that the power of a lens is equal to the reciprocal of its focal length. Therefore, we can write the power of the combination of lenses as:

$$P = 1/f = 1/f_1 + 1/f_2 - d/f_1f_2$$

This is the formula for the focal length of a combination of lenses, taking into account the distance between the two lenses. It can be verified by using the lens maker's formula and the formula for the power of a thin lens in air.

5. To measure attenuation in an optical fiber:

Attenuation is the reduction in the intensity of light as it travels through an optical fiber. It is measured in decibels per kilometer (dB/km). Here are the steps to measure attenuation in an optical fiber:

1. Set up a light source and a power meter: The light source should be connected to one end of the optical fiber, and the power meter should be connected to the other end.
2. Measure the input power: Turn on the light source and measure the power of the light entering the optical fiber using the power meter.
3. Measure the output power: Measure the power of the light exiting the optical fiber using the power meter.
4. Calculate the attenuation: The attenuation can be calculated using the formula: Attenuation (dB) = $10 * \log_{10} (P_{in} / P_{out})$, where P_{in} is the input power and P_{out} is the output power.
5. Repeat the measurement: To ensure accuracy, the measurement should be repeated several times and the average value should be taken.

It is important to note that the attenuation of an optical fiber can vary with the wavelength of the light, so the measurement should be performed at the wavelength of interest. Additionally, the connectors and splices in the optical fiber can also contribute to the attenuation, so they should be taken into account when measuring the attenuation.

6. To determine the wavelength of He-Ne laser light using single slit diffraction.

1. The experiment involves shining a He-Ne laser beam through a single slit and observing the diffraction pattern produced on a screen.
2. The distance between the central maximum and the first minimum of the diffraction pattern is measured.
3. The slit width and the distance between the slit and the screen are also measured.
4. Using the formula for single slit diffraction, the wavelength of the He-Ne laser light can be calculated.

5. The formula for single slit diffraction is: $\lambda = (a \sin \theta) / m$, where λ is the wavelength of the light, a is the width of the slit, θ is the angle of diffraction, and m is the order of the minimum.
6. By substituting the measured values into the formula, the wavelength of the He-Ne laser light can be determined.

7. To study the polarization of light using He-Ne laser light.

1. **Introduction:** Polarization is a property of light that describes the direction of its electric field. Light can be polarized in different ways, such as linear, circular, or elliptical polarization.
2. **He-Ne laser light:** A He-Ne laser is a type of gas laser that uses a mixture of helium and neon gases as the lasing medium. The light emitted by a He-Ne laser is typically linearly polarized.
3. **Polarization of light:** When light passes through a polarizing material, only the component of the electric field that is parallel to the polarization axis of the material is transmitted. The other component is absorbed or reflected.
4. **Experiment:** To study the polarization of light using a He-Ne laser, one can set up an experiment with a polarizer and an analyzer. The polarizer is placed in the path of the laser light to polarize it, and the analyzer is used to measure the intensity of the transmitted light.
5. **Observations:** By rotating the analyzer and measuring the intensity of the transmitted light, one can observe how the intensity changes with the angle between the polarization axis of the polarizer and the analyzer. This can provide information about the polarization state of the laser light.
6. **Conclusion:** The study of the polarization of light using a He-Ne laser can provide valuable insights into the properties of light and its interaction with matter. It is an important topic in the field of optics and photonics.

8. To determine the wavelength of sodium light with the help of Fresnel's bi-prism.

Fresnel's bi-prism is an optical instrument used to determine the wavelength of light. It consists of two prisms placed base to base, forming a bi-prism. When a beam of light passes through the bi-prism, it is split into two beams, which interfere with each other to produce an interference pattern.

To determine the wavelength of sodium light using Fresnel's bi-prism, the following steps can be followed:

1. Set up the bi-prism experiment by placing the bi-prism in the path of a collimated beam of sodium light.
2. Adjust the position of the bi-prism and the eyepiece to obtain a clear interference pattern.

3. Measure the distance between the central bright fringe and a fringe of known order.
4. Calculate the fringe width using the formula: fringe width = (distance between fringes) / (order of fringe).
5. Use the formula for the wavelength of light in a bi-prism experiment to calculate the wavelength of sodium light: wavelength = (fringe width * distance between slit and eyepiece) / (distance between virtual sources).

By following these steps, the wavelength of sodium light can be accurately determined using Fresnel's bi-prism. It is important to note that the accuracy of the measurement depends on the precision of the measurements and the quality of the experimental setup.

9. To determine the coefficient of viscosity of a given liquid.

Viscosity is a measure of a fluid's resistance to flow. The coefficient of viscosity is a measure of the internal friction of a fluid. There are several methods to determine the coefficient of viscosity of a given liquid, including:

1. **Ostwald viscometer:** This method involves measuring the time it takes for a fixed volume of liquid to flow through a narrow tube under the influence of gravity. The coefficient of viscosity can be calculated using the Poiseuille equation.
2. **Falling sphere viscometer:** This method involves measuring the terminal velocity of a sphere falling through a liquid. The coefficient of viscosity can be calculated using Stokes' law.
3. **Rotational viscometer:** This method involves measuring the torque required to rotate a spindle at a constant speed in a liquid. The coefficient of viscosity can be calculated using the relationship between torque, spindle speed, and the dimensions of the spindle and liquid container.

Each of these methods has its own advantages and limitations, and the choice of method may depend on the specific properties of the liquid being tested and the precision required for the measurement. It is important to carefully follow the experimental procedure and to take multiple measurements to ensure accurate results.

10. To determine the value of acceleration due to gravity (g) using compound pendulum.

1. The compound pendulum is a rigid body swinging in a vertical plane about a horizontal axis passing through the body.
2. The time period of a compound pendulum is given by the formula $T = 2\pi\sqrt{I/mgd}$, where I is the moment of inertia of the pendulum about the axis of rotation, m is the mass of the pendulum, g is the acceleration due to gravity, and d is the distance between the center of mass of the pendulum and the axis of rotation.

3. To determine the value of g using a compound pendulum, the time period of the pendulum is measured for different values of d .
4. A graph is plotted between T^2 and d , and the slope of the graph is used to calculate the value of g .
5. The slope of the graph is equal to $4^2(I/mg)$, so g can be calculated as $g = 4^2(I/m \cdot \text{slope})$.
6. The value of g obtained using this method is an average value for the location where the experiment is performed.
7. This method of determining the value of g is useful because it can be performed using simple equipment and does not require advanced technology.
8. The accuracy of the result depends on the accuracy of the measurements of the time period, the mass of the pendulum, and the distance d .
9. The experiment can be repeated multiple times to obtain an average value for g and to reduce the uncertainty in the result.
10. The value of g determined using this method can be compared with the standard value of g to check the accuracy of the result.

Group B

Group B is a term that can refer to many different things depending on the context. Here are some possible interpretations of the term “Group B”:

1. In the context of rally racing, Group B was a set of regulations introduced by the Fédération Internationale de l'Automobile (FIA) in 1982 for competition vehicles in rallying and sports car racing. Group B cars were known for their power, speed, and lack of restrictions, which led to many accidents and fatalities, ultimately resulting in the cancellation of the Group B category in 1986.
2. In the context of microbiology, Group B Streptococcus (GBS) is a type of bacteria that can cause illness in people of all ages. In newborns, GBS can cause sepsis, pneumonia, and meningitis. In adults, GBS can cause urinary tract infections, skin infections, and other illnesses.
3. In the context of vitamins, Group B refers to the B-complex vitamins, which include thiamine (B1), riboflavin (B2), niacin (B3), pantothenic acid (B5), pyridoxine (B6), biotin (B7), folic acid (B9), and cobalamin (B12). These vitamins play important roles in metabolism, energy production, and the maintenance of healthy skin, hair, and nails.

1. To determine the energy band gap of a given semiconductor material:

- The energy band gap of a semiconductor material is the difference in energy between the top of the valence band and the bottom of the conduction band.
- The band gap is a crucial property of a semiconductor material, as it determines its electrical conductivity and optical properties.

- There are several methods to determine the energy band gap of a semiconductor material, including optical absorption, photoluminescence, and electrical measurements.
- In the optical absorption method, the absorption coefficient of the material is measured as a function of the photon energy. The absorption edge, where the absorption coefficient increases sharply, corresponds to the energy band gap of the material.
- In the photoluminescence method, the material is excited with light of a certain energy, and the emitted light is analyzed to determine the energy band gap.
- In electrical measurements, the current-voltage characteristics of the material are measured at different temperatures, and the energy band gap is determined from the temperature dependence of the current.
- The choice of method depends on the specific properties of the material and the experimental conditions.

2. To study Hall effect and determine Hall coefficient, carrier density and mobility of a

- The **Hall effect** is defined as the difference in voltage generated across a current-carrying conductor, is transverse to an electrical current in the conductor and an applied magnetic field perpendicular to the current.
- The drift velocity acquired in unit applied electric field is known as the **mobility** of the carrier and is denoted by μ_H and is also called **Hall mobility**.
- By measuring the resistivity of the materials and knowing the Hall coefficient, density along y-axis and current density along x-axis, we can determine the **Hall angle**.
- The **Hall coefficient** (R_H) and **mobility** (μ_H) are independent of magnetic and electric fields, however they changes with film thickness and composition of the film.
- The **carrier concentration** (n_H) depends on thickness and composition of the films.
- The **Hall measurement**, carried out in the presence of a magnetic field, yields the sheet carrier density n_s and the bulk carrier density n or p (for n -type or p -type material) if the conducting layer thickness of the sample is known.
- The **Hall voltage** for thick, heavily doped samples can be quite small (of the order of microvolts).

Given Semiconductor Material Using Hall Effect Setup

The Hall effect is a phenomenon that occurs in a conductor when an electric current flows through it in the presence of a magnetic field. The magnetic field causes the electrons in the conductor to deflect, creating a voltage difference across the conductor, perpendicular to the direction of the current flow. This

voltage difference is known as the Hall voltage.

The Hall effect can be used to determine several properties of a given semiconductor material, including:

1. **Carrier concentration:** The Hall voltage is directly proportional to the carrier concentration in the semiconductor material. By measuring the Hall voltage, the carrier concentration can be calculated.
2. **Type of semiconductor:** The sign of the Hall voltage can be used to determine whether the semiconductor material is a p-type or an n-type semiconductor. A positive Hall voltage indicates a p-type semiconductor, while a negative Hall voltage indicates an n-type semiconductor.
3. **Mobility:** The Hall voltage is also related to the mobility of the charge carriers in the semiconductor material. By measuring the Hall voltage and the carrier concentration, the mobility of the charge carriers can be calculated.

A Hall effect setup typically consists of a sample of the semiconductor material, a source of a magnetic field, and a means of measuring the Hall voltage. The sample is placed in the magnetic field, and a current is passed through it. The Hall voltage is then measured across the sample, perpendicular to the direction of the current flow.

In summary, the Hall effect can be used to determine several important properties of a given semiconductor material, including the carrier concentration, the type of semiconductor, and the mobility of the charge carriers. A Hall effect setup typically consists of a sample of the semiconductor material, a source of a magnetic field, and a means of measuring the Hall voltage.

3. To determine the variation of magnetic field with the distance along the axis of a current

The magnetic field produced by a current-carrying wire can be determined using the Biot-Savart law. This law states that the magnetic field at a point due to a current element is directly proportional to the current, the length of the current element, and the sine of the angle between the current element and the line joining the point to the current element, and inversely proportional to the square of the distance between the point and the current element.

To determine the variation of the magnetic field with distance along the axis of a current-carrying wire, the following steps can be followed:

1. Set up the apparatus, including a current-carrying wire, a magnetic field sensor, and a measuring device to measure the distance from the wire.
2. Turn on the current and measure the magnetic field at various distances along the axis of the wire.
3. Record the data and plot a graph of the magnetic field strength against the distance from the wire.

4. Analyze the graph to determine the relationship between the magnetic field strength and the distance from the wire.

The relationship between the magnetic field strength and the distance from the wire can be determined by fitting a curve to the data points. The shape of the curve will depend on the specific experimental setup and the strength of the current in the wire. In general, the magnetic field strength is expected to decrease as the distance from the wire increases. This relationship can be used to predict the magnetic field strength at other distances along the axis of the wire.

Carrying Coil and Estimate the Radius of the Coil

- A carrying coil is a coil of wire that carries an electric current.
- The magnetic field produced by the current in the coil can be used to estimate the radius of the coil.
- To estimate the radius of the coil, the magnetic field at the center of the coil can be measured using a Hall probe or a magnetometer.
- The magnetic field at the center of the coil is given by the formula $B = \frac{\mu_0 NI}{2R}$, where B is the magnetic field, μ_0 is the permeability of free space, N is the number of turns in the coil, I is the current in the coil, and R is the radius of the coil.
- By measuring the magnetic field at the center of the coil and knowing the other parameters, the radius of the coil can be calculated using the formula $R = \frac{\mu_0 NI}{2B}$.
- It is important to note that this method of estimating the radius of the coil is only accurate if the coil is tightly wound and the current is uniformly distributed throughout the coil.

4. To verify Stefan's law by electric method.

Stefan's law states that the total energy radiated per unit surface area of a black body in unit time is directly proportional to the fourth power of the absolute temperature of the black body. The law can be verified by the electric method, which involves the following steps:

1. A black body is heated by passing an electric current through it, and its temperature is measured using a thermometer.
2. The temperature of the black body is gradually increased by increasing the electric current, and the corresponding increase in the energy radiated by the black body is measured using a radiometer.
3. A graph is plotted between the fourth power of the absolute temperature of the black body and the energy radiated by it.
4. If the graph is a straight line passing through the origin, it confirms that the energy radiated by the black body is directly proportional to the fourth power of its absolute temperature, thus verifying Stefan's law.

5. To determine resistance per unit length and specific resistance of a given resistance

- Resistance is a measure of the opposition to the flow of electric current through a conductor.
- The resistance of a conductor depends on its length, cross-sectional area, and the material it is made of.
- The resistance per unit length is the resistance of a conductor divided by its length. It is usually measured in ohms per meter (Ω/m).
- The specific resistance or resistivity of a material is a measure of the material's inherent resistance to the flow of electric current. It is usually measured in ohm-meters (Ωm).
- To determine the resistance per unit length and specific resistance of a given resistance, the following steps can be followed:
 1. Measure the length of the conductor using a ruler or measuring tape.
 2. Measure the resistance of the conductor using an ohmmeter.
 3. Divide the resistance by the length of the conductor to obtain the resistance per unit length.
 4. To determine the specific resistance, the cross-sectional area of the conductor must also be measured. This can be done by measuring the diameter of the conductor and using the formula for the area of a circle (πr^2 , where r is the radius of the conductor).
 5. The specific resistance can then be calculated by multiplying the resistance per unit length by the cross-sectional area of the conductor.

Using Carey Foster's Bridge

Carey Foster's Bridge is a method used to measure the resistance of a wire or to compare two resistances. The bridge consists of four resistances connected in a Wheatstone bridge configuration. Here are the steps to use Carey Foster's Bridge:

1. Connect the unknown resistance and the standard resistance in the two gaps of the bridge.
2. Connect a galvanometer between the junctions of the two pairs of resistances.
3. Connect a battery or other source of electromotive force (EMF) between the other two junctions of the bridge.
4. Adjust the sliding contact on the resistance wire until the galvanometer shows no deflection, indicating that the bridge is balanced.
5. Use the known values of the standard resistance and the position of the sliding contact to calculate the value of the unknown resistance.

Carey Foster's Bridge is a precise and accurate method for measuring resistance and is commonly used in laboratory settings. It is important to ensure that the bridge is properly balanced and that the connections are secure to obtain accurate results.

6. To study the resonance condition of a series LCR circuit.

A series LCR circuit consists of an inductor (L), a capacitor (C), and a resistor (R) connected in series. The resonance condition of a series LCR circuit occurs when the inductive reactance (XL) is equal to the capacitive reactance (XC). This results in the circuit having a minimum impedance and maximum current flow.

The resonance frequency (fr) of a series LCR circuit can be calculated using the formula:

$$f_r = 1 / (2 * \sqrt{LC})$$

At resonance, the voltage across the inductor and capacitor are equal in magnitude but opposite in phase. The voltage across the resistor is in phase with the current.

The quality factor (Q) of a series LCR circuit is a measure of the sharpness of the resonance and is given by the formula:

$$Q = (2 f_r L) / R$$

The bandwidth (Δf) of a series LCR circuit is the range of frequencies over which the circuit exhibits resonance behavior and is given by the formula:

$$\Delta f = f_r / Q$$

In summary, the resonance condition of a series LCR circuit occurs when the inductive reactance is equal to the capacitive reactance, resulting in a minimum impedance and maximum current flow. The resonance frequency, quality factor, and bandwidth can be calculated using the formulas provided above.

7. To determine the electrochemical equivalent (ECE) of copper.

1. The electrochemical equivalent (ECE) of a substance is the mass of the substance deposited or dissolved at an electrode during electrolysis by the passage of one coulomb of electric charge.
2. The ECE of copper can be determined experimentally by performing an electrolysis experiment using a copper sulfate solution and copper electrodes.
3. The mass of the copper electrode is measured before and after the experiment to determine the change in mass.
4. The electric charge passed through the solution during the experiment is measured using an ammeter and a stopwatch.
5. The ECE of copper is calculated by dividing the change in mass of the copper electrode by the electric charge passed through the solution.
6. The ECE of copper can also be calculated theoretically using Faraday's laws of electrolysis and the atomic weight of copper.
7. The experimental and theoretical values of the ECE of copper can be compared to evaluate the accuracy of the experiment.

8. To calibrate the given ammeter and voltmeter by potentiometer.

Calibration is the process of comparing the readings of an instrument with a known standard to determine the accuracy of the instrument. In this experiment, we will calibrate an ammeter and a voltmeter using a potentiometer.

The steps to calibrate the given ammeter and voltmeter by potentiometer are as follows:

1. Set up the circuit as shown in the diagram, with the ammeter, voltmeter, and potentiometer connected in series.
2. Adjust the potentiometer to obtain a known current through the circuit.
3. Note the reading on the ammeter and compare it with the known current.
4. If the ammeter reading is not equal to the known current, adjust the zero error of the ammeter accordingly.
5. Repeat the above steps for different values of current to obtain a calibration curve for the ammeter.
6. Similarly, adjust the potentiometer to obtain a known voltage across the voltmeter.
7. Note the reading on the voltmeter and compare it with the known voltage.
8. If the voltmeter reading is not equal to the known voltage, adjust the zero error of the voltmeter accordingly.
9. Repeat the above steps for different values of voltage to obtain a calibration curve for the voltmeter.

By following these steps, the given ammeter and voltmeter can be calibrated using a potentiometer. This will ensure that the readings obtained from these instruments are accurate and reliable.

9. To draw hysteresis (B-H curve) of a specimen in the form of a transformer and to

Hysteresis is the dependence of the state of a system on its history. In the context of magnetic materials, hysteresis refers to the lagging of the magnetization of a material behind the applied magnetic field. This phenomenon is graphically represented by the B-H curve, also known as the hysteresis loop.

To draw the hysteresis loop of a specimen in the form of a transformer, the following steps can be followed:

1. Set up the transformer with the specimen as the core, and connect it to an AC power source.
2. Connect an ammeter in series with the primary coil to measure the current, and a voltmeter across the secondary coil to measure the induced voltage.
3. Vary the current in the primary coil and record the corresponding values of the induced voltage in the secondary coil.
4. Plot the values of the induced voltage (B) against the current in the primary coil (H) to obtain the B-H curve.

The shape of the hysteresis loop depends on the magnetic properties of the specimen. Materials with a large hysteresis loop, such as hard magnetic materials, have a high coercivity and retain their magnetization even after the applied magnetic field is removed. On the other hand, materials with a small hysteresis loop, such as soft magnetic materials, have a low coercivity and lose their magnetization quickly when the applied magnetic field is removed.

Hysteresis Loss

Hysteresis loss is the energy loss that occurs in a magnetic material due to the reversal of magnetization. It is a type of energy loss that occurs in ferromagnetic materials when they are subjected to a changing magnetic field. The energy loss is due to the movement of the magnetic domains within the material.

To determine the hysteresis loss, the following steps can be taken:

1. Measure the area of the hysteresis loop: The area of the hysteresis loop is a measure of the energy loss per cycle. The larger the area of the loop, the greater the energy loss.
2. Calculate the energy loss per cycle: The energy loss per cycle can be calculated by multiplying the area of the hysteresis loop by the frequency of the magnetic field.
3. Determine the total energy loss: The total energy loss can be calculated by multiplying the energy loss per cycle by the number of cycles.

In summary, to determine the hysteresis loss, one can measure the area of the hysteresis loop, calculate the energy loss per cycle, and then determine the total energy loss by multiplying the energy loss per cycle by the number of cycles.

10. To measure high resistance by leakage method.

The leakage method is a technique used to measure high resistance values. This method is based on the principle of measuring the leakage current flowing through the insulation of a conductor. The following are the steps involved in measuring high resistance by the leakage method:

1. The conductor whose resistance is to be measured is connected to a high voltage source.
2. The leakage current flowing through the insulation of the conductor is measured using a sensitive ammeter.
3. The resistance value is calculated by dividing the applied voltage by the measured leakage current.

This method is commonly used to measure the insulation resistance of electrical equipment and cables. It is a simple and effective way to determine the quality of insulation and to detect any defects or deterioration in the insulation material. However, it is important to note that the accuracy of the measurement can be affected by factors such as temperature, humidity, and the presence of

contaminants on the surface of the insulation. Therefore, it is important to take these factors into consideration when using the leakage method to measure high resistance values.

Course outcomes:

1. A course outcome is a statement that describes the knowledge, skills, and abilities that students are expected to have acquired by the end of a course.
2. Course outcomes are used to assess the effectiveness of a course in achieving its intended goals.
3. Course outcomes are typically developed by the instructor and are based on the course content and objectives.
4. Course outcomes should be specific, measurable, and achievable.
5. Course outcomes should be aligned with the overall program outcomes and institutional goals.
6. Course outcomes can be used to guide the development of course content, assessments, and teaching strategies.
7. Course outcomes can also be used to communicate expectations to students and to provide a basis for evaluating student learning.
8. Course outcomes can be assessed through a variety of methods, including exams, assignments, projects, and other forms of assessment.
9. Course outcomes should be regularly reviewed and revised to ensure that they remain relevant and up-to-date.
10. Course outcomes are an important tool for ensuring that students are achieving the intended learning outcomes of a course and for improving the quality of education.

At the end of the course, students will be able to:

- Demonstrate knowledge and understanding of the subject matter.
- Apply the concepts and principles learned in the course to solve problems.
- Analyze and evaluate information critically.
- Communicate effectively in written and oral forms.
- Work collaboratively with others.
- Demonstrate ethical and professional behavior.
- Use technology effectively to enhance learning.
- Engage in lifelong learning and professional development.

CO CO Statement Bloom's

CO CO Statement Bloom's is a topic that refers to the use of Bloom's Taxonomy in the creation of Course Outcome (CO) statements. Bloom's Taxonomy is a framework for categorizing educational goals and objectives into different levels of complexity and specificity. It is often used in the development of assessments and curriculum design.

Here are some key points to consider when using Bloom's Taxonomy to create CO statements:

1. Bloom's Taxonomy consists of six levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating. Each level represents a different type of cognitive process and requires a different level of understanding and skill.
2. When creating CO statements, it is important to consider the level of Bloom's Taxonomy that is most appropriate for the course and the learning objectives. For example, a course that focuses on basic knowledge and understanding may have CO statements that align with the Remembering and Understanding levels, while a course that focuses on critical thinking and problem-solving may have CO statements that align with the Analyzing and Evaluating levels.
3. CO statements should be specific, measurable, and achievable. They should clearly define what the student is expected to know or be able to do by the end of the course.
4. It is important to align the CO statements with the course content, assessments, and instructional strategies. This helps to ensure that the course is designed to effectively support student learning and achievement of the COs.
5. Bloom's Taxonomy can also be used to design assessments that accurately measure student achievement of the COs. Assessments can be designed to target different levels of Bloom's Taxonomy, allowing instructors to assess student understanding and skill at different levels of complexity.

In summary, CO CO Statement Bloom's refers to the use of Bloom's Taxonomy in the creation of Course Outcome statements. This framework can be a useful tool for designing courses and assessments that effectively support student learning and achievement.

Level

- A level is a tool used to determine if a surface is horizontal (level) or vertical (plumb).
- It consists of a small glass tube containing a liquid, usually alcohol or a colored spirit, with an air bubble.
- The tube is mounted in a frame with a flat base, and when the base is placed on a surface, the bubble will move to the highest point of the tube.
- If the bubble is centered between two marks on the tube, the surface is level or plumb.
- Levels come in various sizes and shapes, including torpedo, I-beam, and box-beam levels.
- They are commonly used in construction, carpentry, and surveying, among other fields.

- Laser levels and digital levels are more advanced types of levels that can provide greater accuracy and ease of use.

CO-1 Apply the principle of interference and diffraction to find the wavelength of

- Interference and diffraction are two phenomena that can be used to determine the wavelength of light.
- Interference occurs when two or more waves overlap and combine to form a new wave. The resulting wave can have a larger or smaller amplitude than the original waves, depending on the relative phases of the waves.
- Diffraction is the bending of waves around obstacles or through openings. When light passes through a narrow slit, it spreads out in a pattern called a diffraction pattern.
- The wavelength of light can be determined by measuring the spacing of the fringes in an interference or diffraction pattern.
- One common method for measuring the wavelength of light is to use a double-slit experiment. In this experiment, light passes through two narrow slits and forms an interference pattern on a screen.
- The spacing of the fringes in the interference pattern is related to the wavelength of the light and the distance between the slits. By measuring the fringe spacing and the slit separation, the wavelength of the light can be calculated.
- Another method for measuring the wavelength of light is to use a diffraction grating. A diffraction grating is a surface with many closely spaced parallel lines. When light is shone on the grating, it diffracts and forms a pattern of bright and dark spots on a screen.
- The spacing of the spots in the diffraction pattern is related to the wavelength of the light and the spacing of the lines on the grating. By measuring the spot spacing and the line spacing, the wavelength of the light can be calculated.
- These methods can be used to accurately determine the wavelength of light and are commonly used in scientific experiments and measurements.

Monochromatic and Polychromatic Light

- **Monochromatic light** refers to light that consists of a single wavelength or color. This type of light is produced by lasers, LEDs, and certain types of lamps.
- **Polychromatic light**, on the other hand, consists of multiple wavelengths or colors. This type of light is produced by most light sources, including the sun, incandescent bulbs, and fluorescent lamps.
- The difference between monochromatic and polychromatic light is important in various fields, including optics, spectroscopy, and colorimetry.

- In optics, monochromatic light is used to study the behavior of light as it passes through various optical elements, such as lenses and prisms. This is because monochromatic light is not subject to chromatic aberration, which can distort the image produced by an optical system.
- In spectroscopy, monochromatic light is used to study the interaction between light and matter. By shining monochromatic light on a sample and measuring the light that is absorbed, reflected, or emitted, scientists can determine the chemical composition of the sample.
- In colorimetry, polychromatic light is used to study the color of objects. By measuring the amount of light of different wavelengths that is reflected or transmitted by an object, scientists can determine the color of the object.
- In summary, monochromatic light consists of a single wavelength or color, while polychromatic light consists of multiple wavelengths or colors. The difference between these two types of light is important in various fields, including optics, spectroscopy, and colorimetry.

CO-2 Compute and analyze various electrical and electronic properties of a given

1. To compute and analyze various electrical and electronic properties of a given, one must first understand the basic concepts of electricity and electronics.
2. Electrical properties refer to the behavior of electric charges and electric fields, while electronic properties refer to the behavior of electrons in materials.
3. Some common electrical properties include resistance, capacitance, and inductance. These properties can be measured using various instruments such as an ohmmeter, a capacitance meter, and an inductance meter.
4. Electronic properties include conductivity, mobility, and carrier concentration. These properties can be measured using techniques such as the Hall effect and the four-point probe method.
5. To analyze these properties, one must have a good understanding of the underlying physics and mathematics. This includes knowledge of circuit analysis, electromagnetism, and semiconductor physics.
6. With this knowledge, one can compute and analyze the electrical and electronic properties of a given and make informed decisions about its design and operation.

Material Analysis by Using Various Experiments

Material analysis is the process of determining the chemical composition, structure, and properties of a material. This information is crucial for understanding the behavior of materials in different environments and for developing new materials with improved properties. There are several experiments that can be used to analyze materials, including:

1. **X-ray diffraction (XRD):** This technique is used to determine the crystal structure of a material. It involves shining X-rays onto a sample and measuring the diffraction pattern produced by the X-rays interacting with the atoms in the material.
2. **Scanning electron microscopy (SEM):** This technique is used to image the surface of a material at high magnification. It involves scanning a focused beam of electrons across the surface of the sample and detecting the secondary electrons emitted by the sample.
3. **Transmission electron microscopy (TEM):** This technique is used to image the internal structure of a material at high resolution. It involves passing a beam of electrons through a thin sample and detecting the electrons that are transmitted through the sample.
4. **Energy-dispersive X-ray spectroscopy (EDX):** This technique is used to determine the elemental composition of a material. It involves exciting the atoms in the sample with an electron beam and measuring the characteristic X-rays emitted by the atoms.
5. **Fourier transform infrared spectroscopy (FTIR):** This technique is used to determine the chemical composition of a material. It involves shining infrared light onto a sample and measuring the absorption of the light by the chemical bonds in the material.

These are just a few of the many experiments that can be used to analyze materials. The choice of experiment depends on the specific information that is needed about the material. By using a combination of these techniques, it is possible to gain a comprehensive understanding of the properties and behavior of a material.

CO-3 Verify different established laws with the help of optical and electrical

There are several established laws in the field of optics and electricity that can be verified through experiments and observations. Some of these laws include:

1. **Ohm's Law:** This law states that the current flowing through a conductor is directly proportional to the voltage applied across it, provided the temperature remains constant. This law can be verified by measuring the current and voltage across a conductor and plotting a graph to show the linear relationship between the two.
2. **Snell's Law:** This law describes the relationship between the angles of incidence and refraction when light passes through a boundary between two different media. This law can be verified by measuring the angles of incidence and refraction when light passes through a transparent material such as glass or water.

3. **Kirchhoff's Laws:** These laws describe the relationship between the currents and voltages in an electrical circuit. Kirchhoff's Current Law states that the sum of the currents entering a node in a circuit is equal to the sum of the currents leaving the node. Kirchhoff's Voltage Law states that the sum of the voltage drops around a closed loop in a circuit is equal to the sum of the voltage rises. These laws can be verified by measuring the currents and voltages in a circuit and checking that they satisfy the equations.
4. **Law of Reflection:** This law states that the angle of incidence is equal to the angle of reflection when light is reflected from a smooth surface. This law can be verified by measuring the angles of incidence and reflection when light is reflected from a mirror or other smooth surface.

These are just a few examples of the many established laws in the field of optics and electricity that can be verified through experiments and observations. By verifying these laws, we can gain a better understanding of the fundamental principles that govern the behavior of light and electricity.

Experiments

An experiment is a scientific procedure undertaken to make a discovery, test a hypothesis, or demonstrate a known fact. Here are some key points to consider when conducting an experiment:

1. **Hypothesis:** A hypothesis is a proposed explanation for a phenomenon. It is important to have a clear and testable hypothesis before conducting an experiment.
2. **Variables:** Variables are factors that can be changed or controlled in an experiment. It is important to identify and control variables to ensure the accuracy of the results.
3. **Control group:** A control group is a group in an experiment that does not receive the treatment being tested. The control group is used as a baseline to compare the results of the treatment group.
4. **Data collection:** Data collection is the process of gathering and measuring information. It is important to have a clear and consistent method for collecting data to ensure the accuracy of the results.
5. **Data analysis:** Data analysis is the process of examining, cleaning, transforming, and modeling data to discover useful information. It is important to analyze the data collected in an experiment to draw conclusions and make decisions.
6. **Conclusion:** A conclusion is a summary of the results of an experiment. It is important to draw a clear and accurate conclusion based on the data collected and analyzed.

7. **Replication:** Replication is the process of repeating an experiment to confirm the results. It is important to replicate experiments to ensure the accuracy and reliability of the results.

Apply

- To apply means to make a formal request, usually in writing, for something such as a job, a loan, permission for something, or admission to a school.
- When applying for something, it is important to follow the instructions and provide all the required information.
- An application usually includes personal information, such as name and contact details, as well as information relevant to the request, such as qualifications and experience for a job application.
- It is important to present the information in a clear and organized manner, and to check for errors before submitting the application.
- After submitting an application, it is common to receive a response within a certain timeframe, which may vary depending on the nature of the application.
- If the application is successful, the next steps may include an interview or further communication to finalize the process.
- If the application is unsuccessful, it is usually possible to request feedback to understand the reasons and improve future applications.

CO-4 Determine and calculate various physical properties of a given material by using

1. **Density:** The density of a material is defined as its mass per unit volume. It can be calculated by dividing the mass of the material by its volume.
2. **Elasticity:** Elasticity is the ability of a material to return to its original shape after being deformed. The modulus of elasticity, also known as Young's modulus, is a measure of the stiffness of a material and can be calculated by dividing the stress by the strain.
3. **Thermal conductivity:** Thermal conductivity is a measure of a material's ability to conduct heat. It can be calculated by dividing the heat flow rate by the temperature gradient and the cross-sectional area of the material.
4. **Electrical conductivity:** Electrical conductivity is a measure of a material's ability to conduct electricity. It can be calculated by dividing the current by the electric field and the cross-sectional area of the material.
5. **Melting point:** The melting point of a material is the temperature at which it changes from a solid to a liquid. It can be determined experimentally by heating the material and observing the temperature at which it melts.
6. **Hardness:** Hardness is a measure of a material's resistance to deformation or indentation. It can be determined using various hardness tests, such as the Rockwell, Brinell, or Vickers tests.

These are some of the physical properties of a material that can be determined and calculated using various methods and techniques. It is important to understand these properties in order to select the appropriate material for a given application.

Various Experiments

There are numerous experiments that have been conducted in various fields of science. Some of the most notable experiments include:

1. **The Double-Slit Experiment:** This experiment, first performed by Thomas Young in 1801, demonstrated the wave-like behavior of light. It involved shining a beam of light through two narrow slits and observing the resulting interference pattern on a screen behind the slits.
2. **The Michelson-Morley Experiment:** This experiment, conducted by Albert Michelson and Edward Morley in 1887, was designed to detect the presence of the hypothetical “luminiferous ether” that was thought to permeate all space and serve as the medium through which light waves traveled. The experiment failed to detect any such ether, providing evidence for the theory of relativity.
3. **The Rutherford Gold Foil Experiment:** This experiment, conducted by Ernest Rutherford in 1911, involved firing alpha particles at a thin sheet of gold foil. The results showed that most of the particles passed straight through the foil, while a small number were deflected at large angles. This led Rutherford to propose the existence of a small, dense nucleus at the center of the atom.
4. **The Stanford Prison Experiment:** This experiment, conducted by Philip Zimbardo in 1971, involved assigning college students to play the roles of prisoners and guards in a simulated prison environment. The experiment was intended to study the psychological effects of perceived power, but had to be terminated early due to the abusive behavior of the “guards” and the distress of the “prisoners.”
5. **The Milgram Experiment:** This experiment, conducted by Stanley Milgram in 1961, involved instructing participants to administer electric shocks to a “learner” (who was actually a confederate of the experimenter) whenever the learner made a mistake on a memory task. The experiment was designed to study obedience to authority, and the results showed that a majority of participants were willing to administer what they believed to be potentially lethal shocks when instructed to do so by an authority figure.

These are just a few examples of the many experiments that have been conducted in the pursuit of scientific knowledge. Each experiment has contributed to our understanding of the world around us and has helped to shape the course of scientific inquiry.

Apply

- The word “apply” can have several meanings depending on the context in which it is used.
- It can mean to make a formal request, usually in writing, for something such as a job, a loan, or admission to a school or college.
- It can also mean to put or spread something such as paint, cream, or glue onto a surface.
- Another meaning of “apply” is to use something such as a rule, system, or skill in a particular situation or for a particular purpose.
- In mathematics, “apply” can mean to use a mathematical operation or formula to a set of numbers or other mathematical objects.
- In general, “apply” means to use something in a practical or relevant way.

CO-5 Study and estimate the performance and parameter of given equipment by using

1. To study and estimate the performance and parameter of given equipment, it is important to first understand the specifications and capabilities of the equipment.
2. This can be done by referring to the equipment’s manual or technical documentation, which should provide detailed information on its performance and parameters.
3. Another way to estimate the performance and parameter of the equipment is by conducting tests and experiments. This can involve measuring the equipment’s output and comparing it to its expected performance.
4. Data analysis and statistical methods can also be used to estimate the performance and parameter of the equipment. This can involve analyzing the data collected from the tests and experiments to identify trends and patterns.
5. It is important to note that the accuracy of the estimates will depend on the quality of the data collected and the methods used to analyze it. Therefore, it is important to ensure that the data collection and analysis methods are reliable and appropriate for the equipment being studied.

Graphical and Computational Analysis

Graphical and computational analysis are two methods used to analyze data and solve problems. Here are some key points to consider:

1. **Graphical analysis** involves the use of graphs and charts to visually represent data and identify trends, patterns, and relationships. This method is useful for quickly understanding large amounts of data and making predictions based on the observed trends.
2. **Computational analysis** involves the use of mathematical algorithms and computer programs to analyze data and solve problems. This method

is useful for handling complex calculations and large data sets, and can provide more precise and detailed results than graphical analysis.

3. Both methods have their strengths and weaknesses, and the choice of method depends on the specific problem at hand. Graphical analysis is often used for exploratory data analysis, while computational analysis is used for more detailed and rigorous analysis.
4. In many cases, a combination of both methods can provide the most comprehensive and accurate results. For example, graphical analysis can be used to identify trends and patterns in the data, while computational analysis can be used to test hypotheses and make predictions based on the observed trends.
5. It is important to carefully choose the appropriate method for the specific problem at hand, and to use the results of the analysis to inform decision-making and problem-solving.

Apply

1. The word “apply” can have several meanings, depending on the context in which it is used.
2. It can mean to make a formal request, usually in writing, for something such as a job, a loan, or admission to a school or university.
3. It can also mean to put or spread something onto a surface, such as applying paint to a wall or applying lotion to the skin.
4. Another meaning of “apply” is to use or make use of something, such as applying knowledge or skills to solve a problem.
5. In mathematics, “apply” can mean to use a mathematical operation or formula to a given set of numbers or variables.
6. In general, “apply” refers to the act of putting something into action or practice, or making use of something for a specific purpose.

Reference Books

- Reference books are a type of non-fiction book that provides factual information on a particular subject.
- They are often used for research and academic purposes.
- Some common types of reference books include dictionaries, encyclopedias, atlases, and almanacs.
- Reference books are typically organized in a way that makes it easy to find the information you are looking for.
- They are often updated regularly to ensure that the information they contain is accurate and up-to-date.
- Reference books can be found in libraries, bookstores, and online.
- They are an important resource for students, researchers, and anyone seeking to learn more about a particular subject.

1. Practical Physics- K. K. Dey & B. N. Dutta (Kalyani Publishers New Delhi)

- “Practical Physics” is a book written by K. K. Dey and B. N. Dutta.
- It is published by Kalyani Publishers, New Delhi.
- The book is a comprehensive guide to practical physics experiments and measurements.
- It covers a wide range of topics, including mechanics, heat, optics, electricity, and magnetism.
- The book is written in a clear and concise manner, with numerous diagrams and illustrations to aid understanding.
- It is a valuable resource for students studying physics at the undergraduate level.
- The book provides detailed instructions for performing experiments and taking measurements, as well as guidance on how to analyze and interpret data.
- It is a useful reference for both students and instructors.

2. Engineering Physics-Theory and Practical- Katiyar & Pandey (Wiley India)

- “Engineering Physics-Theory and Practical” is a book written by Katiyar & Pandey and published by Wiley India.
- The book covers the fundamental concepts of physics and their applications in engineering.
- It is designed to provide a comprehensive understanding of the subject for students pursuing engineering courses.
- The book is divided into two parts: Theory and Practical.
- The Theory part covers topics such as mechanics, waves and oscillations, optics, electromagnetism, and modern physics.
- The Practical part includes experiments and activities that help students apply the theoretical concepts in a practical setting.
- The book is written in a clear and concise manner, making it easy for students to understand the subject matter.
- It includes numerous examples, illustrations, and exercises to help students learn and practice the concepts.
- Overall, “Engineering Physics-Theory and Practical” by Katiyar & Pandey is a valuable resource for students studying engineering physics.

3. Engineering Physics Practical- S K Gupta (Krishna Prakashan Meerut)

- “Engineering Physics Practical” is a book written by S K Gupta and published by Krishna Prakashan Meerut.
- The book is intended for students studying engineering and covers practical aspects of physics.

- It includes experiments and practical applications of physics concepts relevant to engineering.
- The book is designed to help students develop practical skills and gain hands-on experience in applying physics to engineering problems.
- It is a valuable resource for students preparing for exams and for those seeking to deepen their understanding of physics in an engineering context.